Dual-Role IoT Monitoring System

Enterprise-Scale Role-Based Dashboard Architecture

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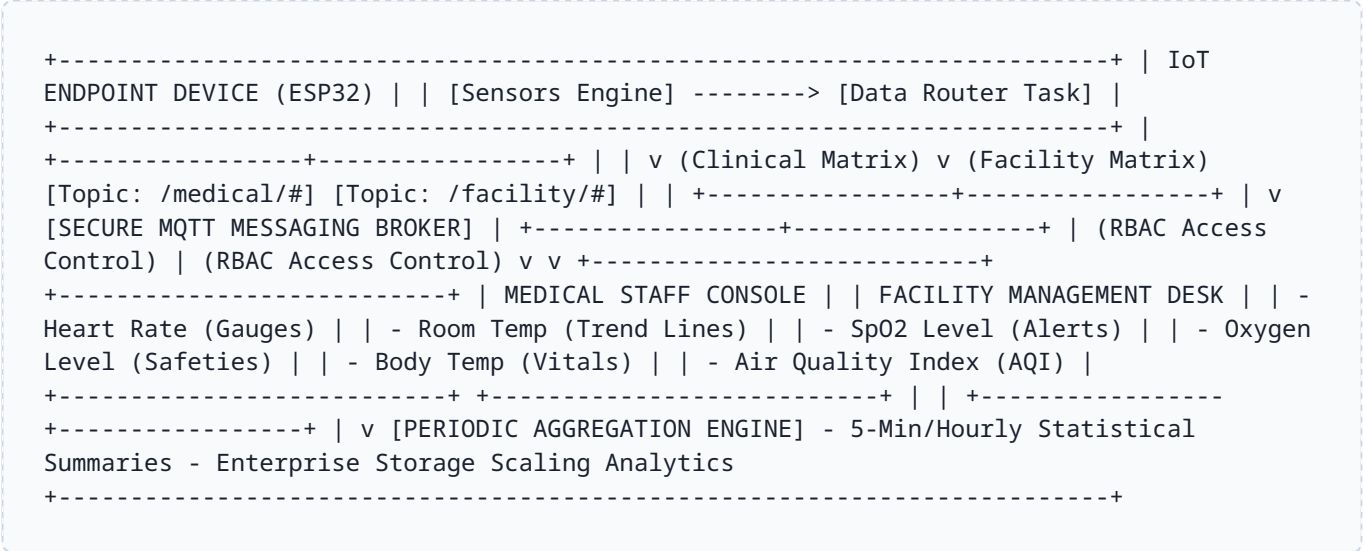
1. System Architecture Diagram

This system architecture separates data pipelines right at ingestion to isolate critical clinical telemetry from facility maintenance parameters. By segregating the messaging topics structurally, the architecture natively provides role-based access control, high security, and minimal endpoint processor footprint.



2. FreeRTOS Task Structure Explanation

To enforce concurrent operation on the edge micro-hardware, the device isolates processing pipelines into distinct tasks utilizing FreeRTOS priority scheduling. Vitals reading is given a higher priority profile to maintain data accuracy during intense processing cycles.

Task Matrix & Resource Controls

Task Handle Name	Priority	Interval / Strategy	Core Architecture Responsibility
vTaskClinicalRead	High (4)	20ms (Periodic)	Polls life-critical biological metrics (SPO2 registers, Heart Rate variations, Temperature data).
vTaskFacilityRead	Medium (2)	2000ms (Periodic)	Monitors environmental safety registers (Ambient Temp, Oxygen partial pressures, AQI data).
vTaskDataRouter	High (3)	Event-Driven	Validates structural boundaries and marshals payloads into respective MQTT routing streams.
vTaskAggregation	Low (1)	300s (5-Min Window)	Computes local sliding mathematical metrics (Averages, Max/Min peaks) for data reduction.
vTaskWatchdog	Highest (5)	100ms (Periodic)	Verifies task health loop checkpoints; refreshes systemic countdown elements.

Resource Synchronization

- **Mutex Bus Separation:** Explicit Mutex flags separate telemetry sensor reading configurations from environmental channels when communicating on common communication paths.
- **Data Serialization Queues:** Buffers data updates dynamically, ensuring network dropped frames do not block execution lines.

3. Feed Mapping & RBAC Routing Table

Data namespaces are explicitly separated into role-specific structures to enforce Role-Based Access Control (RBAC) definitions at the cloud routing layer.

MQTT Target Topic Namespace	Payload Formatting	Authorized Role Profile	Dashboard Widget Elements & Constraints
`/medical/heart_rate`	Integer (BPM)	Clinical / Medical Staff Only	Radial Gauge [Safe Range: 60 - 100 BPM]
`/medical/spo2`	Integer (Percentage)	Clinical / Medical Staff Only	Linear Level Bar [Critical Threshold: < 95%]
`/medical/body_temp`	Float (Celsius)	Clinical / Medical Staff Only	Thermal Gauge Block [Safe Range: 36.1 - 37.2°C]
`/facility/room_temp`	Float (Celsius)	Facility Support / Admin	Historical Trend Line Chart [Range: 20 - 24°C]
`/facility/o2_level`	Float (Percentage)	Facility Support / Admin	Safety Status Indicator [Safe Boundary: > 19.5%]
`/facility/aqi`	Integer (Index Value)	Facility Support / Admin	Color-coded Numeric Counter [Threshold Alert: > 100]

4. Dashboard Viewports & Layout Definition

Role authorization splits the physical representation layer cleanly into two independent system viewpoints.

[Viewport A: Medical Staff Console Interface]

• Heart Rate Interface Widget: Radial Gauge tracking high-frequency pulse inputs.

• SpO2 Monitoring Console: Safety tracking element with automatic blink alert toggles on critical dips.

• Core Patient Temperature Window: Direct telemetry tracker for medical response groups.

[Viewport B: Facility Management Dashboard Desk]

• Environment Tracking Block: Multi-line historical graph showing temperature changes.

• Atmospheric O2 Sensor Module: Static bar monitoring room oxygen concentration safety bounds.

• Air Quality Processing Analytics: Operational index counter displaying ambient dust/gas factors.

5. Safe Threshold Logic & Alert Feeds

To avoid false positives while keeping response times fast, safe boundary logic checking is managed inside edge processor routines before network dispatch:

```
// Boundary Safety Evaluation Structure Execution
void evaluate_clinical_boundaries(PatientData_t data) {
    if (data.spo2 < 95) {
        publish_emergency_alert("/medical/alerts/spo2", "CRITICAL_SPO2_DROP");
    }
    if (data.heartRate > 100 || data.heartRate < 60) {
        publish_emergency_alert("/medical/alerts/pulse", "ANOMALOUS_HEART_RATE");
    }
}
```

Anomalous alerts trigger priority publishes directly to isolated alert channels (e.g., `/medical/alerts/spo2`), forcing real-time sound alarms on corresponding dashboard clients.

6. Periodic Data Aggregation Engine

To operate at enterprise efficiency, the architecture compresses high-frequency streams before pushing records to database pipelines. This reduces cloud network storage overhead and lowers execution data traffic.

Within every 5-minute sliding analysis block, raw data vectors are processed mathematically into statistical aggregates:

$$Vitals_{Avg} = rac{1}{N} \sum_{i=1}^{N} ext{RawValue}_i \quad ext{and} \quad Vitals_{Peak} = \max(ext{RawValue}_1, ext{RawValue}_2, \dots, ext{RawValue}_N)$$

Compressed historical JSON strings (containing average, peak, and low parameters) are published cleanly to specific data warehousing feeds (e.g., `/medical/summary`), creating a streamlined data storage structure.

7. Enterprise Scaling Implementation

The layout replicates production-grade IoT systems through three main mechanisms:

- **Strict Namespace Partitioning:** Uses structural wildcards (such as subscribing to `/medical/+/alerts`) to allow seamless vertical scaling as new hospital wings or sensor channels are added.
- **Broker-Level RBAC Enforcement:** Implements token authorization profiles directly on the central broker, preventing facility personnel from accessing private medical data tracks.
- **Network Conservation Strategies:** Combining edge aggregation with dynamic anomaly triggers significantly minimizes network bandwidth usage, allowing thousands of terminal edge devices to share backend architectures efficiently.